

1. (8 points) Complete lines 5 and 7 in the following function.

```
1 def weeks_and_days(days):
2     """Converts any number of days into weeks plus extra days.
3     For example, if the input is 30, it should print '4 week(s) and 2 day(s)'.
4     """
5     weeks =
6
7     days =
8
9     print(weeks, "week(s) and", days, "day(s).")
```

2. (12 points) For each of the following Python expressions, write down the output when the expression is evaluated using a Python interpreter. Write **error** if you think the expression will raise an error.

(a) `19 % 5`

(b) `5 + 2 * 10`

(c) `"HSC" * 3`

(d) `str(1) + int(2)`

(e) `13 // 4`

(f) `1 + 2 = 3`

3. (8 points) What will the following function return when you call it? Be sure to give both the return value and its type.

```
1 def mystery():
2     x = 5.0
3     y = int(x)
4     z = y ** 2
5     return z - x
```

4. (12 points) Write a loop with an accumulator variable that takes a string `s` and creates a new string where every letter or character in `s` is repeated twice. For example, if `s = "Hello!"`, your loop should generate the string `"HHeellllloo!!"`.

5. (10 points) Identify the type (str, int, float, bool, or list) of each expression.

(a) ["hello", "how", "are", "you?"]

(b) 6 / 2

(c) 5 < 6

(d) -1 + 4 // 5

(e) "4"

6. (8 points) Consider the following function.

```
1 def f(x):  
2     while x > 3:  
3         x = x + 1  
4     return x
```

(a) What will happen if you call f(5)?

(b) What will happen if you call f(2)?

7. (12 points) A list is *increasing* if every element in the list is greater than the element before it. The following function is supposed to return True if a list is increasing and False otherwise, but it has a bug.

```
1 def is_increasing(values):  
2     for i in range(1, len(values)):  
3         if values[i-1] < values[i]:  
4             return True  
5     else:  
6         return False
```

(a) What will is_increasing([2, 3, 5]) return?

(b) What will is_increasing([4, 8, 1, 5]) return?

(c) What is wrong with the way the function is_increasing is coded?

8. (4 points) What will the following code output?

```
1 x = [1, 2, 3, 4]  
2 print(x[2])
```

9. (8 points) Consider the following recursive function which accepts an integer input.

```
1 | def recurse(n):  
2 |     if n <= 1:  
3 |         return 0  
4 |     else:  
5 |         return 1 + recurse(n // 2)
```

(a) What does `recurse(-1)` return?

(b) What does `recurse(10)` return?

10. (8 points) Determine the values of the following boolean expressions.

(a) `5 < 6 and 2 + 2 == 4`

(b) `print(4 + 2) == 6`

(c) `"Fish" in "All the fish in the sea."`

(d) `not (5 > 6 or 1+1 != 3)`

11. (10 points) Trace the following function by making a table to show how the values of the variables `t`, `x`, and `n` change when you compute `result([1, 5, 4])`. Be sure to indicate what the function returns at the end.

```
1 | def result(numbers):  
2 |     t = 0  
3 |     for x in numbers:  
4 |         n = 2 * x  
5 |         t = t + n - 1  
6 |     return t
```